Fiscal Year:	FY 2023	**Task Last Updated:**	FY 02/01/2023
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	Foam Optics and Mechanics		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities
Start Date:	02/20/2019	**End Date:**	03/25/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 03/25/2023 per NSSC information. (Ed. 11/24/21) NOTE: End date changed to 11/30/2020 per NSSC information (Ed., 9/9/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0599		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Continuation of "FOAM--Foam Optics and Mechanics (The Melting of Aqueous Foams)," grant NNX14AM99G, and previously "The Melting of Aqueous Foams," grant numbers NNX07AP20G ; NNX07AK60G ; NNC04GB61G. The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology. There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the “mechanics” aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies Earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into Advanced Life Support (ALS) technology for NASA's strategic exploration mission. The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability. The value of the FOAM project in terms of NASA’s exploration mission lies with Advanced Life Support. Water recovery and recycling are absolutely central to ALS, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Foams are important for a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam. Relatedly, study of foams brings specific insight into multiphase flow and separation phenomena of direct importance for water recovery and recycling operations.
Task Progress:	With NASA-funded postdoctural researcher (postdoc) Anthony Chieco, we published a peer-review paper in Frontiers in Soft Matter, entitled, "Average evolution and size-topology relations for coarsening 2d dry foams", in collaboration with Professor James P. Sethna from Cornell University Department of Physics. [Ed. Note: See Bibliography.] In this work, I derived several new mathematical identities relating the average rate of change of bubble area and statistical measures of average number of sides of bubbles. These are for two-dimensional dry foams, and the starting assumptions are (1) that they evolve only by gas diffusion between neighboring bubbles according to the classic von Neumann law dAj/dt = k(n-6) where Aj is the area of bubble j and nj is its number of sides, and (2) that it is in self-similar scaling state where side-number distributions are constant. In this work, we also performed careful ground-based experiments to test a few of my new identities. This informs our ability to test if coarsening foams are in a scaling state, which is crucial for the ongoing analysis of International Space Station (ISS) operations (ops) data. Also with Anthony Chieco, as well as other members of my group funded from different sources, we contributed to a manuscript entitled, "Aqueous foams in microgravity, measuring bubble sizes", which has just been accepted for peer-reviewed publication in Comptes Rendus. This paper is a collaborative work with our European Space Agency (ESA)-funded European foam physics colleagues, where we report on several methods we jointly developed for analyzing ISS ops data for bubble size information. This includes Diffuse Transmission Spectroscopy (DTS) data, where the probability for incident photons to diffusely transmit through the sample is analyzed in terms of average bubble size. And it includes three methods for extracting the contact area of surface bubbles from image data. I developed DTS in previous years, and helped guide the ESA team in its implementation. A new twist was to account for leakage of photos from the lateral sides of samples. My group alone developed a novel machine-learning method to segment images (i.e., identify bubbles so that their areas and shapes may be extracted). This was important because prior automated methods have difficulty owing to the polydispersity of bubble sizes and shapes as well as non-uniformity of lighting and camera focus. The basic idea was to train a support vector machine to classify pixels as either inside or outside bubbles. This worked brilliantly, and will be the subject of a future publication. Already our code is available to the public at https:// . Much analysis work is now in progress. Two specific projects are now advanced to the point that publications are in preparation. One regards a simple model, and ground-based experimental tests, for how different non-scaling states approach self-similarity. Another regards a model for effective coarsening exponents that I have constructed based on earlier "border-crossing physics" that a student and I published a couple years ago. This is used to analyze exponents from ISS ops data, particularly in regards to how they vary with increasing liquid content of the foams.
Bibliography Type:	Description: (Last Updated: 06/12/2025)

Abstracts for Journals and Proceedings	Durian DJ. "The coarsening of very wet foams in two and three dimensions." APS March Meeting 2022, March 14-18, 2022, Chicago, Illinois. Bulletin of the American Physical Society. 2022 Mar;67(3):Abstract: W18.00001. https://meetings.aps.org/Meeting/MAR22/Session/W18.1 , Mar-2022
Abstracts for Journals and Proceedings	Chieco AT, Durian DJ. "A generalized size-topology identity for 2-dimensional dry foams." APS March Meeting 2022, March 14-18, 2022, Chicago, Illinois. Bulletin of the American Physical Society. 2022 Mar;67(3):Abstract: W18.00006. https://meetings.aps.org/Meeting/MAR22/Session/W18.6 , Mar-2022
Articles in Peer-reviewed Journals	Pasquet M, Galvani N, Pitois O, Cohen-Addad S, Höhler R, Chieco AT, Dillavou S, Hanlan JM, Durian DJ, Rio E, Salonen A. "Aqueous foams in microgravity, measuring bubble sizes." Comptes Rendus. Mécanique. 2023;351(S2):139-161. http://dx.doi.org/10.5802/crmeca.153 , Nov-2023
Articles in Peer-reviewed Journals	Galvani N, Pasquet M, Mukherjee A, Requier A, Cohen-Addad S, Pitois O, Höhler R, Rio E, Salonen A, Durian DJ, Langevin D. "Hierarchical bubble size distributions in coarsening wet liquid foams." Proc. Nat. Acad. Sci. 2023 Sep 14;120(38):e2306551120. https://doi.org/10.1073/pnas.2306551120 , Sep-2023
Articles in Peer-reviewed Journals	Chieco AT, Durian DJ. "Simply solvable model capturing the approach to statistical self-similarity for the diffusive coarsening of bubbles, droplets, and grains." Phys. Rev. E. 2023 Sep 18;108:034606. https://doi.org/10.1103/PhysRevE.108.034606 , Sep-2023
Articles in Peer-reviewed Journals	Chieco AT, Sethna JT, Durian DJ. "Average evolution and size-topology relations for coarsening 2d dry foams." Frontiers in Soft Matter. 2022 Aug 12; 2:941811. https://doi.org/10.3389/frsfm.2022.941811 , Aug-2022